ESP32-Based Edge Node using TinyML and LoRaWAN for Motor Fault Detection

Muvvala Kalpana
Department of ECM
Vignan's Institute of Information Technology
Duvvada, Visakhapatnam, A.P., India
muvvala.kalpana@gmail.com

Sai Ganesh Dasari
Department of ECM
Vignan's Institute of Information Technology
Duvvada, Visakhapatnam, A.P., India
saiganeshdasari45@gmail.com

Karanam Gayathri
Department of ECM
Vignan's Institute of Information Technology
Duvvada, Visakhapatnam, A.P., India
karanamgayathri26@gmail.com

Hemanth Gonthini
Department of ECM
Vignan's Institute of Information Technology
Duvvada, Visakhapatnam, A.P., India
gonthinihemanth@gmail.com

Ramana Reddy R
Department of ECM
JNTUACE
Kalikiri, A.P., India
profrrreddy@yahoo.co.in

Gembali Dinesh
Member of Technical Staff
Hcl Technologies Pvt. ltd.,
Hyderabad, Telangana, India
gembalidinesh9@gmail.com

Abstract -The failure of induction motors leads to great industrial downtime, but traditional predictive maintenance tends to be too expensive and inaccessible to retrofit. The paper proposes a wireless edge node at a low cost which employs ESP32 and MPU6500 sensor to detect faults in real-time. With TinyML, the model locally receives and classifies motor conditions (healthy, idle, malfunctioning) using vibrations without cloud transfer with high bandwidth requirements. The alerts are sent through LoRaWAN which is long range and ensures connectivity in highly industrial areas. Benchmarking tests show 98.8 correct decisions and a response time of less than 2ms. This solution will validate the hypothesis that Edge AI with LoRaWAN is a powerful and cost-effective method of controlling industries.

Keywords: TinyML, LoRaWAN, Industrial internet of things (IIoT), Predictive Maintenance, Edge Computing, Motor Fault Detection, ESP32.

I.Introduction

The use of industrial induction motors is vital to the industry, where they consume almost 70-80 percent of the energy in the industry. The maintenance strategies are still ineffective though, reactive maintenance results in expensive unplanned downtime, and preventive maintenance usually causes premature parts replacement. Two main challenges in the realization of the modern monitoring system are connectivity and bandwidth. The industrial setting, which is mostly packed with metal constructions, is a hindrance to the normal Wi-Fi signal. Moreover, the conventional IoT strategies send streams of unprocessed vibration data to the cloud, which overloads the bandwidth and eats battery resources of the devices, which is largely unnecessary because the data is mostly redundant. To overcome these shortcomings, this paper presents a low-cost, wireless Edge Node using ESP32 and TinyML. Using an onboard neural network to classify motor health (healthy, idle or faulty) locally, it is possible to switch processing to the edge bypassing the cloud solution. It can be communication of simple status updates over LoRaWAN, a long-range protocol that can penetrate thick industrial fundamentals where Wi-Fi cannot propagate.

Hardware Feasibility & Low-Cost Sensing Foundations of predictive maintenance are established long ago [13], however the recent tendencies are focused on the availability of solutions that are easily accessible and low-cost [12]. Although skepticism early was that the cheapness of sensors could compromise their trustworthiness, recent research shows that MEMS sensors (such as MPU6050/6500) have the required bandwidth to implement industrial fault detection [11], effectively dispelling the argument that inexpensive hardware has excessive noise levels [2]. whereas in terms of processing unit, comparative studies of the Arduino, STM32 and ESP32 show the latter is more optimal in the specified application; the dual-core nature of the ESP32 provides the required tradeoff between speed and power efficiency The ESP32 is capable of inference times as low as 14ms as indicated by benchmarks [4], compared to more expensive industrial components.

Signal Processing Frameworks Raw vibration data are usually too noisy to be directly fed to small neural networks. Therefore, spectral analysis (FFT) is another important pre-processing process to obtain frequency domain features [3]. Although the traditional algorithms such as the Random Forest are computationally light, the Neural Networks have been found to be better at detecting complex, non-linear fault patterns [5]. In order to close the divide between intricate models and limited hardware, we resort to the use of Edge Impulse platform. This makes it easy to have a smooth "Data-to-Deployment" pipeline [6] and allows to have Int8 quantization, retaining high accuracy (>95%) and keeping the limited RAM of ESP32 [1] in mind.

Connection in Industrial Factories It is a major challenge to have reliable data transfer in high density factory environments. Studies confirm that LoRaWAN can work in worst-case scenarios like underground mines [7], which proves that it can be used to penetrate concrete walls that Wi-Fi cannot penetrate. The TinyML and LoRaWAN convergence is being mentioned as a scientifically justified approach to the contemporary IIoT applications [8] which responds to the highly sought-after industry requirement of scalable, long-range monitoring [9].

XXX-X-XXXX-XXXX-X/XX/$XX.00 ©20XX IEEE

1.1 Motivation

As per available literature, the current industrial motor monitoring systems have a number of limitations that are critical. The initial challenge is the high price that would deterrify the majority of the conventional industrial equipment and excessive skepticism concerning the accuracy and bandwidth of low-cost MEMS sensors as far as professional fault detection is concerned. Furthermore, application of Machine Learning to microcontrollers is still rather technical and must be implemented manually, which is usually inefficient and cannot be optimized to high accuracy because of low RAM. Taylor also has problems of getting raw vibration data which is usually very noisy, unstructured to be immediately interpreted by a small neural network, and simpler algorithms like Random Forests not able to capture the less linear, non-linear patterns of motor faults. Additional gaps of deployment and connectivity exist. The edges devices are exposed to memory bottlenecks which are hard to define in more general workflow, and require specific pipelines and quantization techniques. The protocols of low-power architecture like the LoRaWAN with respect to connection in obstructed and harsh industrial systems are usually questionable. Moreover, compared to the case of TinyML and LoRaWAN that are popularly studied independently, there is some literature gap on the hybridization of the two and a lack of quantitative performance benchmarks like ESP32. Finally, the majority of the academic responses are not aligned with the trends in the commercial sector and lack in considering the principles set by the previous literature, which makes them less practical.

1.2 Proposed Work

The proposed system is a method of transforming a regular induction motor into a smart and self-diagnostic device with a combination of the low cost of the hardware and a strong Edge AI. The main part of the design is the ESP32 microcontroller, the processing unit, which will be linked with the MPU6500 accelerator, which will probably be what the system will regard as ears since it reads high-frequency vibration data directly off the motor chassis. Instead of uploading this noisy raw data to a cloud server and inefficiently, the system can process all of it using the TinyML technology and then applies a custom neural network, trained and quantized using Edge Impulse to extract the frequency patterns (FFT) in real-time and then distinguishes between "Normal," "Idle," and Fault" states. Once any fault is detected with high accuracy, the device triggers the use of long-range voice- an SX1278 LoRa module- which does not overload the Wi-Fi network, but transmits a sparse message of change in status over kilometers to a central receiver. The design will lead to an effective and decentralized monitoring system that can offer the performance of deep learning at the power consumption and coverage required in industrial environments that are harsh.

II. Methodology

This section outlines how the proposed IIoT system is to be designed and implemented by replacing the ideal architecture with an experimental model.

2.1 System Design

The architecture proposed makes use of a decentralized Edge AI solution where data is processed on the sensor node instead of on the cloud. The system has two main components; a Transmitter Node (Edge Node) that is mounted on the motor and a remote Receiver Node (Gateway).

A. Transmitter Node (Edge Node) Edge node captures physical vibrations and does on-device inference.

- **Sensing:** MPU6500 6-axis IMU will be used as this sensor has high sensitivity, and minimal noise. It sends the acceleration data (X and Y and Z planes) to the microcontroller through the I2C protocol at 100Hz.
- **Processing unit:** Since ESP32-WROOM-32 is the central processor, it is named the Processing Unit. It implements the TinyML neural network and functions by loading unprocessed vibration data and classifying it as (Normal, Fault, or Idle) motor state.

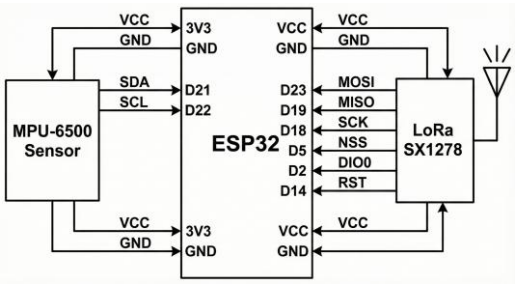


Fig.1. schematic diagram of transmitter end.

As can be seen in Fig. 1, ESP32 is used as the main control board and it will regulate 3.3 V rail to drive all peripherals and maintain the signal integrity. It also can be connected to the sensor in the MPU-6500 through I2C protocols (pin D21, D22) and the LoRa SX1278 module to the standard SPI interface (pin D23, D19, D18, D5) with digital lines utilized on pin D2 and D14 as interrupt and reset functions.

- **Communication Unit:** There is an SX1278 LoRa Module in it which is connected to SPI. When compared to Wi-Fi where one would need to connect Wi-Fi, the LoRa module can be enabled only when the ESP32 has a piece to transmit, which makes it draw the minimum amount of power possible.

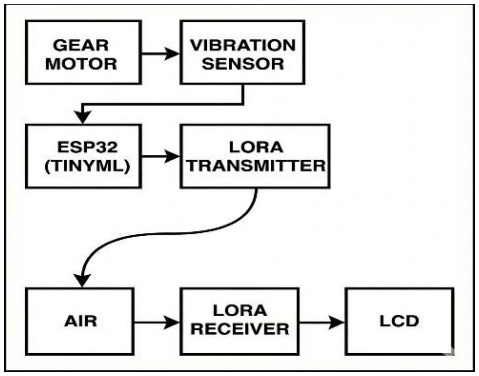


Fig.2.Flowchart

Fig.2 illustrates an entire data flow of both a predictive maintenance or a monitoring system. A LoRa Transmitter sends the data that has been processed and these are eventually shown on a LCD screen in the Air.

B. The Receiver Node (Gateway) This is a receiver node. It consists of a second ESP32, equipped with display LCD, SX1278 LoRa, and has a module. It gets packets that are generated by the edge node and de-serializes the status byte of the packet and displays the operator-real time health status of the motor.

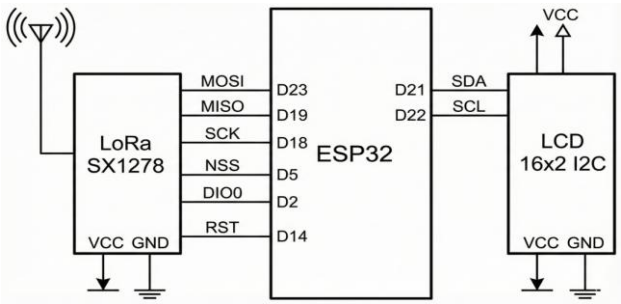


Fig.3. schematic diagram of receiver end

Fig. 3 is shown on the receiver side of the connection and recognizes the signal and transmits the processed data to an LCD display to monitor the signal in real-time.

2.2 Prototype Construction

The physical component involved the inclusion of the physical components in a small, robust, frame which would be in a position to fit in a test motor.

A. Hardware Assembly A prototyping board was used to bring together the prototype so that connections are made to be secure.

1.Power Management: The system has a 5V USB input, which is configured to 3.3V to provide ESP32 and LoRa module. Specific care was taken of the fact that the logic had to be in 3.3V range, and any attempt to subject the LoRa module to 5V would break the transceiver.

2.Sensor Attachment: The MPU6500 sensor was stuck to the chassis of 12V DC motor using industrial adhesive tape to ensure that the transmissibility of vibrations was correct but not damped.

3.Antenna Design: A helical antenna also was soldered to the LoRa module at 433MHz in order to maximize the amount of range the transmission would cover and the termination of the signal past the barriers.

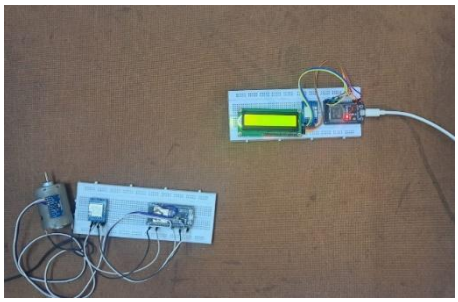


Fig. 4. Proto type model

Fig. 4 shows the predictive maintenance solution is a system architectural design and physical prototype to track the health of a motor with the assistance of edge computing and long-range wireless communication

B. Software and TinyML Pipeline

The implementation was using the Edge Impulse that serves to facilitate machine learning workflow.

1.Information Intake: ESP32 was programmed to store 2 seconds of data in accelerator.

2.Signal Processing: The addition of a Spectral Analysis block into the pipeline was made to make use of a Fast Fourier Transform (FFT). This changes frequency peaks which are distinctive motor speed and fault vibration signals of the time domain vibration signals.

3.Quantization: a 32-bit floating-point trained neural network was quantized to an 8-bit integer (Int8) one. This was a significant move so as to have the model occupying less RAM in ESP32 and yet attain the various rate of inference.



Fig. 5. shows the information in display when fault is detected at node

Fig. 5 describes a predictive maintenance system of monitoring the health of some motor and illustrates a receiver which can put real-time alerts such as ; Motor 1 is in fault condition on the LCD screen.

2.3 Testing and Validation

To start with the reliability of the system, a harsh test regime was carried out to demonstrate the reliability of the system with a test bed of the 12VDC motor.

A. Dataset Collection Custom dataset Double-checking of the induction of conditions on the motor manually:

Normal Operation: the motor worked well and at the rated speed.

Simulated Fault: A blockage was introduced to the rotor shaft to cause an imitated fault that was a simulated error referred to as a mechanical imbalance fault that sustained typical vibration modes.
Idle State: The motor was switched off to get some background noise in the setting. Hundred and fifty minutes of data were vindicated split into training (80 percent) and testing (20 percent) data.

B. TinyML model Model Performance Metrics

Confusion Matrix: The accuracy of the model in the model was 98.8. Under the validation phase, the system obtained false positive zero with the normal and fault states.

Table. 1 confusion matrix

	Fault	*Idle*	*Normal*
Fault	100%	0%	0%
Idle	0%	100%	0%
Nomral	3.7%	0%	96.3%
F1 score	0.98	1.00	0.98

Latency Test: test Results averaged 14milliseconds to gather some information, execute the inference and deliver an outcome, which validated that the system can be utilized to carry out close-to-real-time surveillance.

C. Field Communication Test:

The transmitter was placed 500m above receiver and concrete walls were located in between the distance. The system can send fault alerts at a packet loss rate of less than 1 percent, which is a validation of the assertion of the project as the Long-range valid.

III. System Architecture Proposal

The proposed system in this paper is a decentralized IIoT system that is event-driven. In contrast to cloud-based architectures which receive raw data and stream it to the cloud, this solution employs embedded TinyML to analyze vibration data on the "Edge" and only actionable diagnostic data is sent.

3.1 Hardware Framework (The Edge Node): The physical layer makes use of inexpensive parts that are optimized in terms of power consumption:

1. **Central Processing Unit:** ESP32-WROOM-32 (240MHz dual-core Xtensa(r) LX6) is chosen because it supports milliseconds-level neural network inference and it is able to use deep sleep modes between cycles.
2. **Vibration Measurement:** MPU6500 6-axis MEMS IMU will record acceleration data through its I2C interface (100 Hz, +-2g sensitivity) which has been tuned to read low-to-mid frequency harmonics indicative of bearing wear and unbalance.
3. **Long-range:** Many of these products use the Chirp Spread Spectrum (CSS) modulating the SX1278 LoRa module (433 MHz) to attain multi-kilometer range and penetrate through dense industrial walls where Wi-Fi can no longer operate.

3.2 Algorithm Design (TinyML):

The intelligence pipeline in the Edge Impulse platform was created based on three steps:
Signal Preprocessing (DSP): A Spectral Analysis (FFT) block converts raw time-domain data into the frequency domain to isolate certain fault-related peaks, which may be noise and gravities vectors.
Model Architecture: A Feedforward Neural Network (FNN) computes input and hidden layers with 2 dense layers with ReLU activation and 33 spectral input variables. The last Softmax layer determines the motor state as a normal, idle or fault state.
Quantization: Due to the small amount of RAM that the ESP32 has, the model is quantized to 8 bits of integer (Int8) format. This is showing a reduction of model size by approximately 75% and 98.8 percent accuracy.

3.3 Event-Driven Communication Protocol:

To keep bandwidth and battery consumption down, the system does not support continuous streaming, instead supporting decision-supported logic:
1.Local Inference: ESP32 goes to sleep after each 2 seconds and samples vibrations and conducts a locally-based inference.
2.Decision Logic: When the state is alive or idle, the device will revert to sleep where the device will only transmit a one-hourly heartbeat. In case there is a Fault (confidence value above 0.8), the LoRa module is forced to run.
3.Payload Transmission Unlike in Ethernet, critical alerts are encoded into a 1-byte payload. This guarantees a low airtime of less than 50ms that will greatly decrease the probability of packet collisions and power usage.

IV. Findings and Discussion

The system was tested in two phases, testing the TinyML model on unseen test data and also implementing hardware latency and communication range testing.

4.1 TinyML Model Performance:

The neural network was trained on a 15 minutes vibration data set (80/20 train/test split). The Int8 quantized model was able to classify at 98.8%. The confusion matrix was analyzed and shows that the state of Idile was identified with 100 percent, and sensitivity was 97.5 percent to artificial faults with a low error rate, which was negligible (0.8). The efficiency of resource consumption in the ESP32 was high with the use of 1.4 KB RAM (Peak) and 19.2 KB Flash. The system has been proven to have a real-time monitoring ability without deactivating other parallel processes with an inference time of less than 2 ms.

4.2 Hardware and Communication Metrics:

End-to-end testing was performed upon an 12VDC motor by imposing mechanical imbalance.

System Response Time: Response time was measured as the total time it took between the disruption of the system and the receipt of an alert and was recorded at 2.15 seconds. This time is a sum of the fixed 2-second sampling window of LoRa, inferencing time of approximately 14ms, and the time to transmit payloads over LoRa numbering 50ms.

LoRaWAN Range: Test was done in an active industrial setting that had reinforced concrete and metal buildings. The system was able to provide 0% packet loss using three concrete walls covering a distance of 150 meters (indoor) and provided constant Line-of-Sight (LoS) connectivity to the 1.2 km (outdoor) distance. These findings prove the viability of the system when compared with the traditional Wi-Fi systems which normally experience signal degradation after 50 meters in a similar environment.

V Conclusion and Future Scope

5.1 Conclusion

This paper has introduced a cost effective, wireless edge node to predictive maintenance of the induction motors. The system overcomes the constraints of traditional monitoring: it is expensive, inefficient in use of bandwidth, and has a poor connection, but by combining TinyML with operation based on low-energy ESP32, adding LoRaWAN connectivity to it, it can address those constraints. The experimental findings state that the system is 98.8 percent accurate and the time taken to infer the motor state (Normal, Idle, Fault) is less than 2 ms. In addition, the LoRa technology provided good coverage and reliable transmission of alerts up to more than 1 km hence demonstrated that the solution was feasible in large-scale industrial and agricultural settings. This paper has verified that effective edge intelligence and the use of available embedded technology can be used to conduct high-performance predictive maintenance.

5.2 Future Scope

Future iterations will aim at extending the capabilities of the system on four areas. In the first place, the multi-fault classification will be adopted, which means that the neural network will be trained on a wide range of data to discern particular defects such as cracks in the rotor bars or misalignment. Second, the architecture shall be developed to reveal point to point to LoRaWAN Mesh Network where the nodes will be able to hop data through the broad complex of huge complex without central gateway. Third, energy harvesting solutions (like piezoelectric vibration sensors) will be included in order to allow a deploy and forget autonomy. Lastly, a cloud dashboard will be created that will ease the process of historical information logging, trend analysis, and notifications over a distance through SMS/Email.

VI References

[1] J. P. B. Lima, A. F. R. Trajano, R. B. S. Silva, A. G. Silva-Filho, and P. H. G. Rocha, "Real-Time Fault Detection in Induction Motors Using TinyML: An Evaluation of the Edge Impulse Platform," *IEEE Access*, vol. 11, pp. 12056–12068, 2023. [Online]. Available: https://ieeexplore.ieee.org/abstract/document/10056987

[2] P. Kolok, M. Hodoň, P. Ševčík, L. Hotz, and N. Remy, "Low-Cost IoT-Based Predictive Maintenance Using Vibration and Acoustic Analysis," *Sensors*, vol. 25, no. 21, p. 6610, 2025. [Online]. Available: https://www.mdpi.com/1424-8220/25/21/6610

[3] S. Arciniegas, D. Rivero, J. Piñan, E. Diaz, and F. Rivas, "IoT Device for Detecting Abnormal Vibrations in Motors Using TinyML," *Discover Internet of Things*, vol. 5, no. 1, art. 41, 2025. [Online]. Available: https://link.springer.com/article/10.1007/s43926-025-00142-4

[4] S. Gupta and S. N. Shivhare, "Embedded TinyML for Predictive Maintenance: Vibration Analysis on ESP32 with Real-Time Fault Detection in Industrial Equipment," *International Journal on Computational Modelling Applications*, vol. 2, no. 2, pp. 1–17, 2025. [Online]. Available: https://submissions.adroidjournals.com/index.php/ijcma/article/view/114

[5] G. R. Padule, V. V. Zambare, D. R. Patil, and S. A. Bhosale, "Smart Predictive Maintenance of Induction Motor using ML," *ITSI Transactions on Electrical and Electronics Engineering*, vol. 14, no. 1, pp. 48–52, 2025. [Online]. Available: https://journals.mriindia.com/index.php/itsiteee/article/view/854

[6] S. K. Sahoo, S. S. Sahu, and B. Appasani, "TinyML-Powered Predictive Maintenance Framework on Resource-Constrained Devices: A Blended Learning Approach," in *2025 8th International Conference on Computing Methodologies and Communication (ICCMC)*, 2025, pp. 1–6. [Online]. Available: https://ieeexplore.ieee.org/document/11140652

[7]A. R. Chowdhury, S. C. Bakshi, and A. Pramanik, "Design and Study of LoRa-Based IIoT Network for Underground Coal Mine Environment," *IEEE Access*, vol. 13, pp. 4985–5004, 2025. [Online]. Available: https://ieeexplore.ieee.org/iel8/6287639/10820123/10699321.pdf

[8] M. A. Lodhi, M. S. Obaidat, and L. Wang, "Tiny Machine Learning for Efficient Channel Selection in LoRaWAN," *IEEE Internet of Things Journal*, vol. 11, no. 19, pp. 30714–30724, 2024. [Online]. Available: https://ieeexplore.ieee.org/document/10472403

[9] Y. Abadade, A. Temouden, and H. Bamoumen, "A Comprehensive Survey on TinyML," *IEEE Access*, vol. 11, pp. 96892–96922, 2023. [Online]. Available: https://ieeexplore.ieee.org/document/10177729

[10] A. P. Pangarkar, S. Shabadi, and P. G. Gawande, "A Comprehensive Analysis of ESP32 Microcontroller for IoT Applications," *International Journal of Novel Research and Development (IJNRD)*, vol. 10, no. 10, pp. c499–c507, Oct. 2025. [Online]. Available: https://ijnrd.org/papers/IJNRD2510255.pdf

[11] D. Sackey, A. N. N. Adjei, and J. K. A. Amuzu, "Performance Analysis of Low-Cost MEMS Accelerometers for Industrial Vibration Monitoring," in *2020 IEEE 7th International Conference on Adaptive Science & Technology (ICAST)*, 2020, pp. 1–6. [Online]. Available: https://ieeexplore.ieee.org/document/9245561

[12] Texas Instruments, "Optimizing System Fault Detection in Real-Time Control Systems with Edge AI-Enabled MCUs," Texas Instruments, White Paper SPRY345, 2023. [Online]. Available: https://www.ti.com/lit/wp/spry345/spry345.pdf

[13] S. B. Chaudhury, M. Sengupta, and K. Mukherjee, "Vibration Monitoring of Rotating Machines Using MEMS Accelerometer," *International Journal of Scientific Engineering and Research (IJSER)*, vol. 2, no. 9, pp. 6–11, Sep. 2014. [Online]. Available: https://www.ijser.in/archives/v2i9/SjIwMTMzNTg=.pdf